

SAMRAT MAJHI

AI/ML Engineer | Agentic AI Developer

work.samrat24@gmail.com

+91 6291324785

Samrat.portfolio

github.com/samrat

linkedin.com/in/samrat

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Kolkata, India

SUMMARY

AI/ML Engineer specializing in agentic AI systems — architected a 14-agent LangGraph healthcare assistant (100+ concurrent conversations, 100% grounded RAG responses) and a banking intent classifier at 94% accuracy. Final-year B.Tech CSE student skilled in Python, LangChain, LangGraph, FastAPI, Docker, and AWS/GCP.

SKILLS

Languages & Libraries:

Python, JavaScript, SQL, Numpy, Pandas, Scikit-learn, TensorFlow, Keras, PyTorch, Matplotlib, Seaborn.

Machine Learning & Data Science:

Data Cleaning, Model Training, Hyperparameter Tuning, Data Pipeline, Exploratory Data Analysis, A/B Testing.

GenAI & Deep Learning:

LangChain, LangGraph, RAG, Hugging Face, Vector Database, LLMs, RNN, CNN, LSTM, Neural Networks.

MLOps & Deployment:

Docker, FastAPI, AWS, GCP, CI/CD, MLflow, Git.

PROJECTS

ArogyaAI – Multilingual WhatsApp Multi-Agent Healthcare Assistant

|| [LiveApp](#) || [Code](#)

Technologies Used: Python, JavaScript,CSS, FastAPI, LangGraph, RAG, Twilio, Docker, AWS, EC2, Vector Database.

- Architected and deployed a **14-agent LangGraph** healthcare assistant on **AWS EC2**, handling **100+ concurrent conversations** across symptom triage, follow-up, disease guidance, and emergency escalation via **Twilio WhatsApp**.
- Engineered a RAG pipeline over a curated vaccination knowledge base, **ensuring 100% of responses stayed grounded in verified medical source data** (zero fabricated facts), delivering context-aware multilingual guidance to users.
- Designed an end-to-end **MLOps** architecture using **Docker, FastAPI, AWS EC2, and Vercel**, exposing scalable **REST APIs** and enabling continuous deployment of a production-ready AI healthcare platform.

User Intent Detection & Drift Monitoring System

|| [LiveApp](#) || [Code](#)

Technologies Used: Python, Scikit-learn, TF-IDF, LR, FastAPI, Render, Docker.

- Automated banking query routing with **94% intent classification accuracy**, cutting **manual triage effort by 30%**, by building a **TF-IDF + Logistic Regression** model across balance, loan, fraud, and card queries for downstream **NLP workflows**.
- Reduced misrouted requests by 25%** by engineering a confidence-based drift detection framework that flags **low-confidence predictions (<0.75)** for early model degradation detection.
- Built and deployed a production-ready **MLOps pipeline** with **FastAPI, Streamlit, Docker, Apache Airflow, and GitHub Actions CI/CD**, enabling automated retraining, **live monitoring**, and self-service **issue resolution**.

WORK HISTORY

Pinnacle Labs Pvt Ltd : Data Science Intern

|| 07/2025 - 08/2025 || Remote

- Trained **spaCy + ML classification models** for sentiment analysis and fake news detection, achieving **89% accuracy** on validation datasets across multiple model comparisons.
- Built a **CNN-based image classification pipeline** using **TensorFlow**, with end-to-end data preprocessing, cleaning, and model evaluation — deployed on **GCP**.

Quirky Connections Pvt Ltd : Generative AI Developer Intern

|| 03/2025 - 04/2025 || Remote

- Built and deployed **2 production-ready chatbots** with **PDF summarization and voice support** using **GPT-4/Gemini APIs**, achieving **sub-2-second** response time on **AWS EC2**.
- Containerized with **Docker** and served via **FastAPI** REST backend on **AWS EC2** for reliable production deployment.

EDUCATION

Bachelor of Technology in Computer Science Engineering (MAKAUT)	09/2023 – 07/2026
Pailan College of Management & Technology	CGPA – 8.6 Kolkata
Computer Science and Technology Diploma (WBSCTVESD)	08/2020 – 08/2023
Budge Budge Institute of Technology	CGPA - 7.4 Kolkata

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Data Science
 - AWS Certified Machine Learning - Specialty (MLS-C01)

COMPETITIONS & HACKATHONS

Amazon ML Challenge – 2025: Developed a Smart Product Pricing model by processing 75,000+ product images and training deep learning models on Google Colab for large-scale data handling and optimization.

Google Gen AI Hackathon - 2025: Built a generative AI solution to support and improve youth mental wellness.